

Bank Marketing Campaign

Predictive Dial ML for Term Deposit Subscription

Optimizing marketing efficiency through machine learning to identify high-potential term deposit subscribers and reduce campaign costs.

Presented to: Marketing Director

Dataset: 7,805 customer records | Model: Random Forest | Primary Metric: Recall

Presentation Flow

01

Business Problem

- Term deposit subscription prediction
- and cost-benefit framing

02

Data Overview

- 7,813 customer records across
- 11 features from bank campaigns

03

Exploratory Data Analysis

- Key patterns in customer behavior
- driving deposit decisions

04

Data Preprocessing

- Simple Imputer (Median), Feature engineering, encoding,
- and scaling pipeline design

05

Model Selection

Cross-validated comparison of 8 ML algorithms via Recall

06

Hyperparameter Tuning

Random Forest optimization + threshold calibration at 0.4

07

Model Evaluation

Recall 75%, Precision 65%
Confusion matrix deep-dive

08

Business Impact

38.6% cost reduction —
\$6,656/day saved vs. call-all strategy

09

SHAP Insights

Feature importance and key business drivers explained

10

Conclusion & Recommendations

Tiered outreach strategy and deployment roadmap

Business Problem

Problem Statement

Bank of America conducts telemarketing campaigns to sell term deposit products. The marketing team calls all customers indiscriminately, resulting in:

- High operational costs from unnecessary calls
- Agent fatigue/ineffective and customer frustration
- No prioritization of high-potential leads

The challenge: Build a predictive model that identifies WHICH customers are most likely to subscribe to a term deposit, enabling the bank to focus resources on high-probability leads.

Dataset at a Glance (Balanced)

7,805

Customer
Records

11

Features

48%

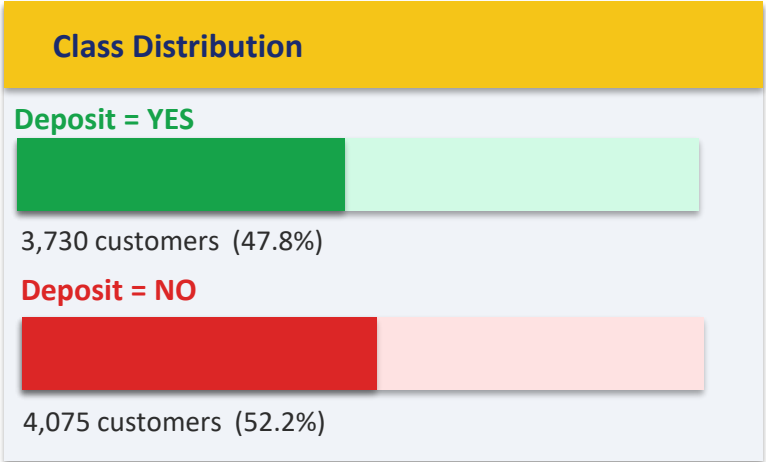
Positive Class
(Deposit = Yes)

Cost-Benefit Framework

- True Positive (TP): Correctly identifies a depositor → Revenue captured
- False Positive (FP): Calls a non-depositor → \$0.1 wasted call cost
- False Negative (FN): MISSES a depositor → \$1 opportunity loss (HIGH COST)
(Source FDIC Quarterly Banking Profile (Q4 2025): An official U.S. government report summarizing the financial performance of 4,336 federally insured banks)
- True Negative (TN): Correctly skips a non-depositor → Cost saved
- → Strategic Priority: MAXIMIZE RECALL to minimize missed depositors

Dataset Overview — Bank Marketing Campaign

Feature Description		
age	Numeric	Age of the customer
job	Categorical	Type of job (admin, technician, services, etc.)
balance	Numeric	Average yearly account balance (EUR)
housing	Binary	Has a housing loan? (yes/no)
loan	Binary	Has a personal loan? (yes/no)
contact	Categorical	Contact communication type (cellular, telephone, unknown)
month	Categorical	Month of last contact
campaign	Numeric	Number of contacts during current campaign
pdays	Numeric	Days since last contact from previous campaign (-1 = not contacted)
poutcome	Categorical	Outcome of previous marketing campaign
deposit ★	Target (Binary)	Did the client subscribe to a term deposit? (yes/no)



Train/Test Split

80% Training → 6,244 rows
20% Testing → 1,561 rows

Stratified split preserves the 48/52 class ratio in both sets.

No data leakage — preprocessing fitted inside each CV fold only.

Key EDA Findings

Contract Type Drives Churn

Month-to-month customers show the highest churn/non-deposit rate. Customers with longer-term prior engagements are significantly more likely to convert.

Insight: Shorter engagement history = lower commitment to new products.

Month of Contact is Significant

Certain months (May, June, July, August) see higher campaign activity but variable success. March, September, October, December show higher conversion rates.

Insight: Campaign timing should align with historically successful months.

Balance Matters

Customers with higher average yearly balances show stronger propensity to deposit. Outliers exist but are valid business values — retained for modeling.

Insight: Wealth-segmentation should inform priority tier assignments.

Previous Campaign Success = Best Predictor

Customers who converted in a previous campaign are far more likely to convert again — strongest signal after threshold calibration.

Insight: Customer retention and repeat engagement programs are high-ROI.

Contact Type is a Predictor

'Unknown' contact type customers are far less likely to deposit. Cellular contact performs best.

Insight: Verified contact channels are critical for campaign effectiveness.

Housing Loan = Negative Indicator

Customers with active housing loans are less likely to deposit — likely due to limited disposable income and existing bank commitments.

Insight: Deprioritize housing-loan customers in outreach tiers.

Preprocessing Pipeline

1 Feature Engineering

- Encode target: yes→1, no→0
- Create was_contacted_before
- Binary flag from pdays
- Replace pdays=-1 with 0

2 Column Transformer

- RobustScaler → numeric cols (handles outliers via IQR)
- OrdinalEncoder → housing, loan (binary yes/no → 0/1)
- OneHotEncoder → job, contact, month, poutcome

3 Train-Test Split

- 80/20 split
- stratify=y to preserve class distribution
- random_state=0 for reproducibility

4 Pipeline Wrapper

- Preprocessing inside Pipeline
- Prevents data leakage in CV
- Fit only on training folds
- Transform applied uniformly

⚠ Multicollinearity Finding

Spearman correlation analysis revealed a high correlation between was_contacted_before and pdays. However, this was intentionally retained because:

- Random Forest and ensemble methods handle multicollinearity naturally via feature diversity at each split.
- Both features carry distinct predictive signal for campaign response.

Why RobustScaler?

balance and campaign contain significant outliers (valid business extremes). RobustScaler uses median + IQR instead of mean + std — making it inherently resistant to extreme values without discarding the outlier data points that carry valuable business information.

Model Selection — 5-Fold Cross-Validation (Recall)

Model	Type	Recall Mean	Recall Std	Status
AdaBoost	Ensemble/Boost	0.643	±0.024	
Random Forest	Ensemble/Bag	0.634	±0.023	✓ SELECTED
Voting	Ensemble/Soft	0.633	±0.027	
KNN	Individual	0.614	±0.026	
Stacking	Ensemble/Meta	0.615	±0.036	
Decision Tree	Individual	0.609	±0.019	
Bagging	Ensemble/Bag	0.592	±0.029	
Logistic Reg.	Individual	0.581	±0.029	

Why Random Forest?

Handles Multicollinearity

Feature diversity at each split neutralizes correlated features naturally.

Robust to Outliers

Averaging mechanism (Bagging) reduces impact of extreme values in balance and campaign.

Overfitting Control

AdaBoost is more susceptible to noise; RF's averaging provides stability.

Class Imbalance

`class_weight='balanced'` parameter handles minority class effectively.

SHAP Compatibility

Tree-based SHAP explainer provides reliable business interpretability.

Hyperparameter Tuning & Threshold Calibration

RandomizedSearchCV Configuration				
n_estimators	[100, 200, 300, 500]	More trees → lower variance		
max_depth	[5, 8, 10, 15, None]	Shallower → less overfitting		
min_samples_leaf	[1, 5, 10, 20]	Larger → smoother boundaries		
min_samples_split	[2, 5, 10, 20]	Fewer splits → less overfit		
max_features	['sqrt', 'log2', 0.5]	Diversity between trees		
class_weight	['balanced', None]	Handles near-imbalance		
Best CV Recall Score	Best n_estimator	Min sample leaf	Max Depth	Class Weight
0.65	300	1	None	Balanced

Recall Improvement After Tuning		
71%	75%	+4%
Base Model Recall (Test)	Tuned Model Recall (Test)	Improvement

Threshold Calibration (0.3 → 0.5)			
Threshold	Recall	Precision	Strategy
0.30	86%	52%	Aggressive
0.35	81%	59%	High Recall
0.40	75%	65%	★ OPTIMAL
0.45	68%	71%	Balanced
0.50	61%	76%	Conservative

Why Threshold = 0.40?

At threshold 0.40, the model achieves the optimal balance:

- Recall = 75% → Captures 3 out of 4 actual depositors
- Precision = 65% → Majority of contacted leads are genuine prospects
- Total daily calls reduced from 7,813 → ~4,296 (45% fewer calls)
- Net cost savings: \$6,656/day compared to calling all customers

Model Evaluation — Final Performance

Recall

75%

75 of every 100 actual depositors are captured

Precision

65%

65% of predicted depositors actually convert

F1-Score

0.70

Balanced harmonic mean of Recall & Precision

Accuracy

69%

Overall correct classifications on test set

Confusion Matrix — Tuned Model (Test Set)

	Predicted: NO	Predicted: YES
Actual: NO	TN 978 Correctly identified non-depositors	FP 302 Called but did not convert
Actual: YES	FN 186 Missed actual depositors ⚠️	TP 559 Correctly identified depositors ✓

Classification Report Summary

Class: No Deposit (0)
Precision: 0.84 | Recall: 0.76 | F1: 0.80

Class: Deposit (1)
Precision: 0.65 | Recall: 0.75 | F1: 0.70

Overfitting Check:
Recall Train = 0.93 | Recall Test = 0.75
Gap = 0.18 — model generalizes reasonably well
with class_weight='balanced' reducing overfit tendency

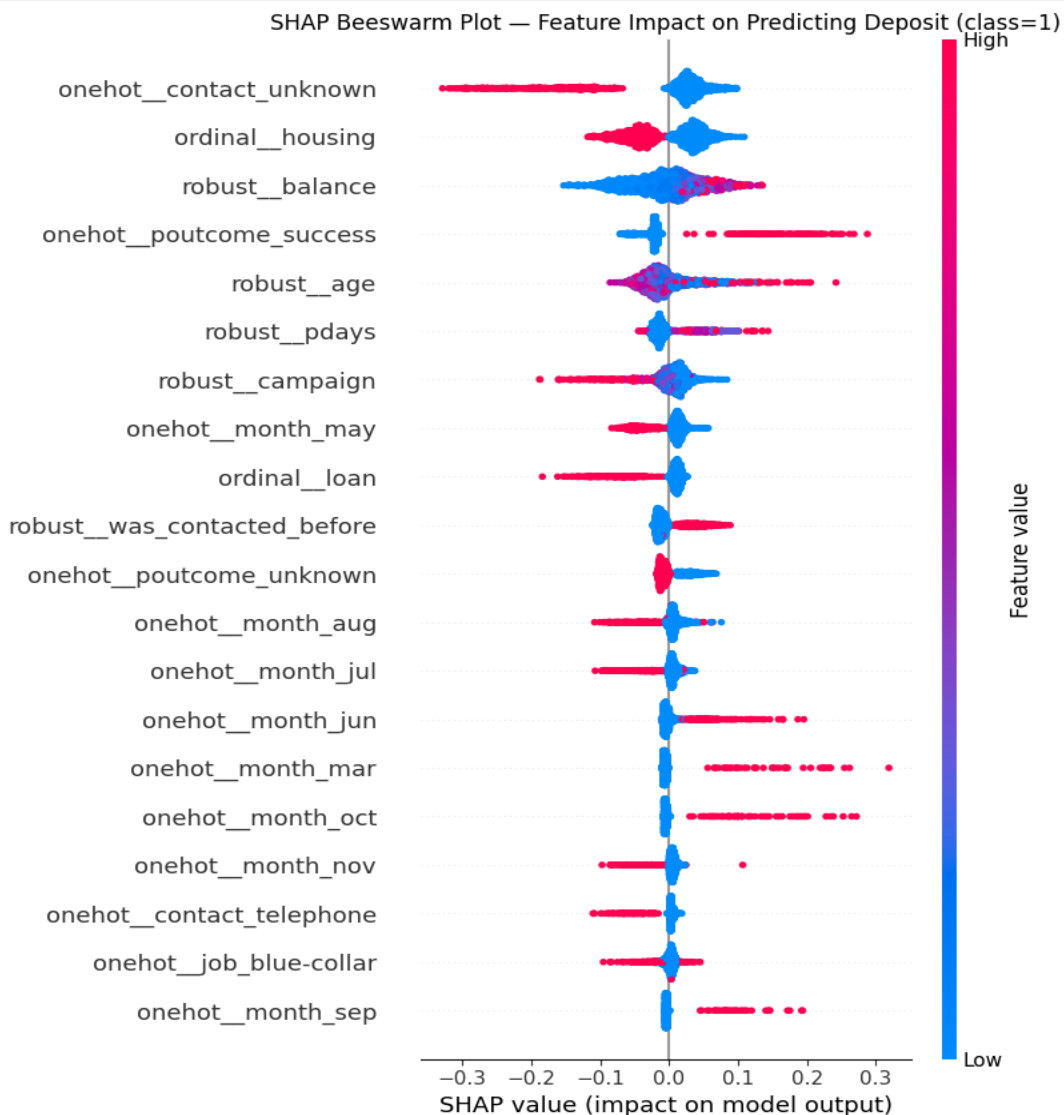
Primary KPI — Test Recall = 75% ✓ Achieved

Business Evaluation — Model vs. No Model

WITHOUT MODEL — Call All Strategy		
Total Calls / Day	7,805	All leads contacted
Depositors Captured	3,732	100% Recall (perfect)
Missed Opportunities	0	No False Negatives
Operational Call Cost	\$781/day	\$0.1 per call × 7,813
Opportunity Loss	\$0/day	All depositors contacted
Staff Required	87 agents	Assuming 4 min/call, 6hr/day
Staff Overhead	\$16,477/day	\$362,500/month
TOTAL DAILY COST	\$17,258/day	High operational burden

WITH MODEL — Targeted Strategy (Threshold 0.40)		
Total Calls / Day	4,296	Precision-filtered leads
Depositors Captured	2,793	75% Recall maintained
Missed Opportunities	939	~\$939 opportunity loss
Operational Call Cost	\$430/day	\$0.1 per call × 4,296
Opportunity Loss	\$939/day	FN × \$1 cost
Staff Required	48 agents	39 staff freed up
Staff Overhead	\$9,091/day	\$200,000/month
TOTAL DAILY COST	\$10,602/day	38.6% more efficient

SHAP & Top 10 Feature Explanation



SHAP Interpretation (Top 5)

📞 Unknown Contact Channel

If the contact method is unknown (red), it significantly **decreases** the chances of a deposit. Known contact methods are crucial for success.

🏠 Housing Loan = Negative

Customers with a housing loan (red) are **less likely** to subscribe to a deposit, likely due to existing financial commitments.

💰 Account Balance

Higher account balances (red) generally lead to a **higher** probability of subscribing to a deposit.

✅ Previous Campaign Success

If the previous marketing campaign was a success (red), there is a very strong **positive impact** on the current prediction.

📅 Age

Older customers (red) show a trend of being **more likely** to deposit compared to younger ones.

SHAP Feature Importance & Business Insights

Top Feature Importance (Mean |SHAP| Value)

Onehot__contact_unknown	0.0606
Ordinal__housing	0.0429
Robust__balance	0.0393
Onehot__poutcome_success	0.0373
Robust__age	0.0292
Robust__pdays	0.0219
Robust__campaign	0.0209
Onehot__month_may	0.0204
Ordinal__loan	0.0201
Robust__was_contacted_before	0.0186

Business Interpretation

Unknown Contact Channel

No 1 Predictor : Customers with 'unknown' contact type are far less likely to deposit. Clean your CRM and verify contact data before campaigns.

Housing Loan = Negative

Customers with housing loans are less likely to subscribe. Deprioritize or segment this group in outreach planning.

Account Balance

Higher-balance customers convert more. Consider a wealth-based segmentation approach for premium tiering.

Previous Campaign Success

If a customer converted before, they are significantly more likely to convert again. Prioritize repeat customers.

Age

Old customer are more likely to subscribe in Deposito program

Summary & Strategic Action Plan

75%

Recall

75 of every 100 depositors
successfully identified

65%

Precision

Majority of leads contacted
are genuine prospects

38.6%

Cost Reduction

\$6,656 saved daily vs.
call-all strategy

39

Staff Freed

High-value reallocation
to other revenue streams

Strategic Recommendations

A

Tiered Outreach Strategy

Segment leads by predict_proba score:
Priority 1 (>0.80): Senior agents, same-day calls
Priority 2 (0.60–0.80): Standard campaign queue
Priority 3 (0.40–0.60): Email/SMS to conserve budget

B

CRM Data Quality Program

Unknown contact type is a major negative predictor. Invest in contact verification to ensure cellular and telephone channels are validated before launch.

C

Seasonal Campaign Scheduling

Focus high-intensity campaigns in March, September, October, and December — historically highest conversion months per SHAP analysis.

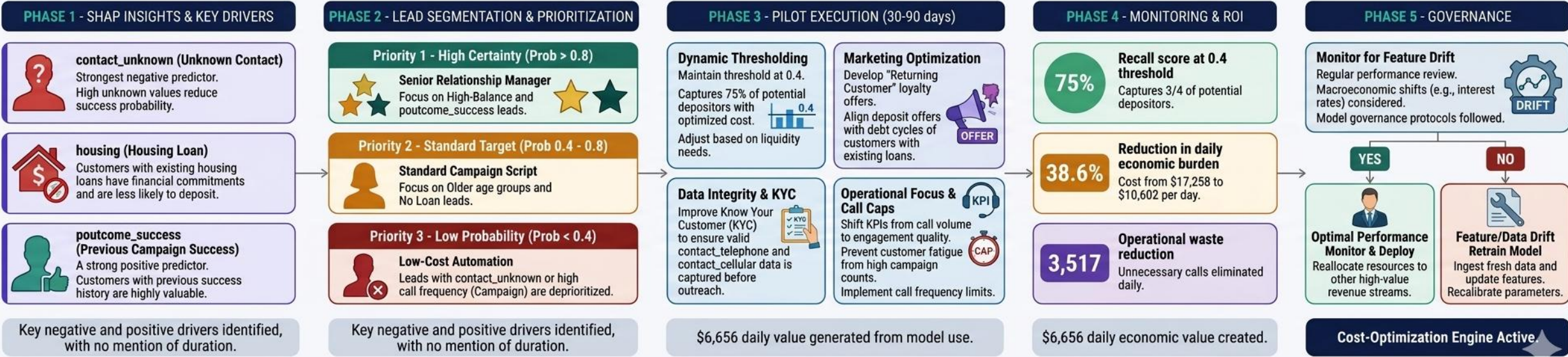
D

Retention & Repeat Engagement

Previous campaign success is the #1 predictor. Build a loyal customer re-engagement program to consistently target prior subscribers first.

Summary & Strategic Action Plan

CONCLUSION AND STRATEGIC RECOMMENDATIONS



Tuned Random Forest Model | Threshold = 0.4 | Recall = 75% | Precision = 65% | 45% reduction in operational workload | 39 skilled staff reallocated